A-A-1200A
May 22, 1989
SUPERCEDING
A-A-1200
April 17, 1980

COMMERCIAL ITEM DESCRIPTION

COATED ABRASIVE CLOTH, SILICON CARBIDE

The General Services Administration has authorized the use of this commercial item description in lieu of Federal Specification P-C-451E.

This commercial item description covers coated abrasives comprised of a backing of cloth coated on one surface with silicon carbide abrasive grain.

Salient characteristics:

Product form. The abrasive cloth, unless otherwise specified, shall be furnished in 9 X 11 (± 1/8 inch dimensional tolerance for width, length is minimum as stated) inch sheet form and of the following grit sizes, as specified:

Grit sizes (Grades) - 320, 280, 240, 220, 180, 150, 120, 100, 80, 60, 50, 40, 36, 30, and 24.

Abrasive. The cloth shall be coated on one side with crushed abrasive quality silicon carbide, graded in accordance with ANSI B74.18 "SPECIFICATIONS FOR GRADING OF CERTAIN ABRASIVE GRAIN ON COATED ABRASIVE PRODUCTS."

Abrasive coverage (as specified):

Open coat - The abrasive grain shall be evenly dispersed over 40 to 70 percent of the backing surface.

Closed coat - The abrasive grain shall completely and uniformly cover the surface of the backing.

Adhesive. The making and sizing coats shall be glue or a comparable adhesive, or synthetic resin. When waterproof abrasive cloth is specified, the adhesive shall be of a good quality drying oil, varnish or other suitable adhesive which will minimize dislodging of the grain from the backing under hard wet or dry usage of the coated product. Bonding of the abrasive grain to the backing shall be equal to or better than that considered good commercial quality.

Backing material. The coated abrasive shall be fabricated from cotton or synthetic cloth, waterproof (when specified), conforming to the following requirements:

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Minimum average coated product breaking strength (lbs/in)		
Cloth type	Warp direction	Filling direction
Jeans ("J")	80	35
Drills ("X")	130	50

Testing.

Grading. The abrasive grain used for the coated abrasive product shall be tested for proper grading in accordance with ANSI B74.18 and the tolerances therein shall apply to the abrasive grain recovered from the coated abrasive product. Failure of the sample grain to conform with grading tolerances shall be cause for rejection of the entire lot.

Test for breaking strength. Minimum average breaking strength shall be determined in accordance with the cutstrip method of ASTM 'STANDARD TEST METHODS FOR BREAKING LOAD AND ELONGATION OF TEXTILE FABRICS', ASTM D1682, using 1 inch strips of the coated abrasive product. Samples shall be tested immediately after being conditioned for 24 consecutive hours at 70° ± 2°F, and 50 ± 2% relative humidity. Failure of the sample strips to conform with stated minimum average requirements shall be cause for rejection of the entire lot.

Test for waterproofness. Samples of abrasive cloth shall be selected from each inspection lot of waterproof product and subjected to the following procedure: After conditioning each sample by submerging in boiling water for a minimum of 15 minutes, subject each sample to the following test. After sufficient cooling to prevent burning of the skin, gently rub each sample between the thumb and forefinger. If the grain coating is dislodged or if the backing cloth tears or is observed to have lost it's body, the sample shall have failed testing.

Identification. The coated abrasive sheets shall be identified by:

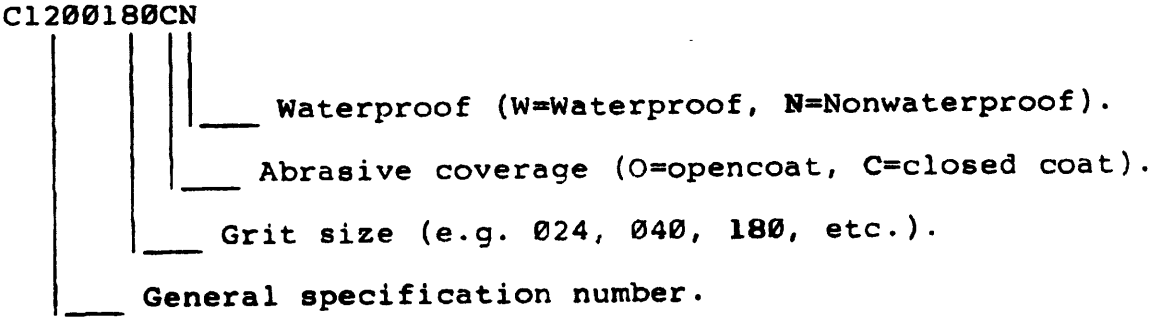
(a) Kind of abrasive grain (generic name, trade name or identification)
(b) Grit size
(c) Backing weight, "X" for drills, "J" for jeans or actual basis weight.
(d) Waterproof designation if applicable.
(e) Open coat identification, if applicable
(f) Manufacturer's name or trademark

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Workmanship. The abrasive sheets shall be free from imperfections which may impair durability or serviceability and shall be equal to or better in every respect than that which is considered good industry practice.

Part Numbering. Coated abrasive cloth conforming with this commercial item description shall be identified by a part number configuration consisting of identification of a portion of the specification number and grit size. An example of the part number configuration is shown below.

EXAMPLE:



Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

Sampling.

Lot. All sheets of the same grade and composition presented for inspection at one time shall be considered a lot for purposes of inspection.

Sampling for visual and dimensional examination. A random sample of sheets shall be selected from each lot in accordance with MIL-STD-105, SAMPLING PROCEDURES AND TABLE FOR INSPECTION BY ATTRIBUTES, inspection level II, AQL 6.5 expressed in terms of defects per hundred unit. The sample unit shall be one sheet.

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Sampling for tests. A random sample of sheets shall be selected from each lot in accordance with MIL-STD-105 special inspection level S-2, the AQL for waterproof testing shall be 2.5 expressed in terms of percent defective. Lot acceptance or rejection for breaking strength shall be based on the average value obtained of the samples tested. The sample unit shall be one sheet.

Frequency of testing. Unless otherwise specified, each bulk roll of coated abrasive product shall be tested for conformance with the abrasive grading and coated product breaking strength requirements specified in this commercial item description. If the end-item product (sheets) cannot be directly traced to the bulk roll by lot or batch number marking applied by the manufacture or by objective documented evidence, then specimens shall be selected from each lot as defined above and tested in accordance with the specified test methods.

Certification of grading, breaking strength and waterproof testing. When specified in the contract or purchase order, a certificate of conformance may be furnished in lieu of actual testing on a lot by lot basis providing that each of following conditions exist:

(1) Objective evidence dated not more than one (1) year prior to the date of the start of the contract as described in the solicitation or date of the purchase order as applicable shall be available.
(2) Tests specified in this document and the industry standards referenced herein have been performed by an independent test facility acceptable to the Government.
(3) That the product being offered is the same type and grade of abrasive and backing material and has been manufactured using the same material, processes and manufacturing techniques as the product tested.

Examination for preparation for delivery. A random sample of unit, intermediate and shipping containers (as applicable) shall be selected from each lot and examined for conformance with the preservation, packaging, packing, labeling and marking required in the contract or order. Samples shall be selected in accordance with MIL-STD-105, special inspection level S-4, AQL 4.0 expressed in terms of percent defective.

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Regulatory requirements. In accordance with section 23.403 of the Federal Acquisition Regulations, the Governments policy is to acquire items composed of the highest percentage of recovered materials practicable, consistent with maintaining a satisfactory level of competition without adversely affecting performance requirements or exposing suppliers' employees to undue hazards from the recovered materials.

Preservation,packaging,packing,labeling,and marking (Civil Agencies). The preservation, packaging, packing, labeling, and marking shall be as specified in the contract or order.

Packaging (DoD). The packaging shall be in accordance with ASTM Standard D3951.

Notes:

Ordering activities should specify:
1) Grit size, backing material (drills or jeans cloth), type of coating (open or closed) and waterproof backing if required.
2) If certification of grading, breaking strength, and practical use testing is permitted in lieu of lot-by-lot testing.

The issue or revision of all documents referenced herein in effect on the date of the invitation for bids or request for quotation shall be used to determine compliance with these requirements.

ANSI standards are available from the American National Standards Institute, 1430 Broadway, New York, NY 10018.

ASTM documents are available from the American Society For Testing And Materials, 1916 Race Street, Philadelphia, PA 19103.

Copies of Military Standard 105 are available from the Naval Publications and Forms Center, 5801 Tabor Ave., Philadelphia, PA 19120.

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MILITARY INTERESTS:

Military Coordinating activity

Navy - YD

Custodians

Army - MR
Navy - YD

Review activity:

Air Force - 82

User:

Navy - AS

PREPARING ACTIVITY

GSA - FSS

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